

TUTONCHANY, Russian Federation

WMO: 235890

Lat: 64.208N

Lon: 93.793E

Elev: 347

StdP: 14.51

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-63.2	-57.3	-66.3	0.1	-62.6	-61.4	0.1	-57.3	15.0	0.1	12.5	-1.5	0.1	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	84.0	63.4	80.2	62.0	76.0	60.5	66.5	77.2	64.2	74.3	62.4	72.4	5.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	85.8	70.5	60.6	80.0	68.2	58.5	74.1	66.2	31.4	76.9	29.6	74.4	28.2	72.0	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.5	11.5	-62.0	89.5	7.7	1.7	-67.6	90.7	-72.1	91.6	-76.4	92.6	-82.1	93.8		
(5)			-61.4	69.7	8.0	2.4	-67.2	71.4	-71.9	72.8	-76.4	74.2	-82.2	75.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.8	-26.6	-13.8	7.6	24.1	37.2	57.0	61.1	54.9	40.4	22.8	-11.7	-17.4	(6)
(7)		DBStd	33.23	21.46	17.36	16.14	10.62	8.19	8.74	5.98	6.92	7.58	12.55	19.10	20.33	(7)
(8)		HDD50	11813	2374	1787	1316	777	404	33	1	35	299	844	1852	2091	(8)
(9)		HDD65	16548	2839	2207	1781	1227	862	263	147	318	737	1309	2302	2556	(9)
(10)		CDD50	797	0	0	0	0	7	245	347	186	12	0	0	0	(10)
(11)		CDD65	56	0	0	0	0	0	24	28	4	0	0	0	0	(11)
(12)		CDH74	1191	0	0	0	0	2	536	553	98	2	0	0	0	(12)
(13)		CDH80	314	0	0	0	0	0	163	140	11	0	0	0	0	(13)
(14)	Wind	WSAvg	4.5	3.1	3.4	4.7	6.2	6.2	5.1	4.6	4.1	4.6	4.7	3.4	3.4	(14)
(15)	Precipitation	PrecAvg	18.7	1.1	0.8	0.7	0.9	1.6	2.3	2.1	2.4	1.9	2.1	1.5	1.2	(15)
(16)		PrecMax	25.9	2.0	1.3	1.9	1.7	2.4	3.9	6.1	4.8	3.1	3.3	2.6	2.2	(16)
(17)		PrecMin	11.7	0.2	0.4	0.2	0.3	0.6	0.4	0.5	1.0	0.8	0.5	0.6	0.4	(17)
(18)		PrecStd	3.3	0.4	0.3	0.4	0.4	0.5	1.1	1.3	1.0	0.7	0.7	0.5	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	30.5	40.3	51.0	71.4	88.8	87.4	81.5	69.9	48.8	31.8	27.5	(19)
(20)			MCWB	17.7	27.6	33.2	39.5	53.7	63.2	66.3	63.1	54.6	43.0	30.0	25.9	(20)
(21)		2%	DB	15.1	20.8	36.9	47.5	63.4	84.9	84.2	76.3	61.4	43.1	26.1	23.8	(21)
(22)			MCWB	13.6	19.0	32.4	37.3	49.4	62.2	64.6	61.4	51.6	39.4	24.7	22.2	(22)
(23)		5%	DB	11.3	16.4	33.5	43.5	56.3	81.0	80.6	72.8	56.0	40.0	23.3	19.0	(23)
(24)			MCWB	10.0	15.0	29.5	35.8	45.1	61.0	63.0	60.3	48.4	37.5	22.3	17.7	(24)
(25)		10%	DB	6.4	12.1	30.6	39.6	50.9	76.4	76.7	68.5	52.2	36.8	17.4	12.3	(25)
(26)			MCWB	5.4	11.0	27.6	33.8	42.0	59.3	61.8	58.7	46.9	34.4	16.1	11.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	27.7	35.3	41.2	55.1	67.9	71.0	66.5	58.3	43.6	29.8	25.9	(27)
(28)			MCD B	19.4	30.5	37.7	47.8	70.5	82.4	82.4	76.5	66.9	48.7	31.8	26.9	(28)
(29)		2%	WB	13.8	19.2	32.3	38.6	50.2	64.8	67.9	63.8	52.7	40.4	25.1	22.2	(29)
(30)			MCD B	15.0	20.3	36.5	45.2	62.6	78.1	78.5	71.8	58.8	42.5	26.2	23.8	(30)
(31)		5%	WB	10.2	15.0	30.0	36.6	46.2	62.7	65.4	61.8	49.7	37.4	22.1	18.1	(31)
(32)			MCD B	11.4	16.4	33.2	42.2	54.6	76.2	74.9	69.7	55.0	39.3	23.4	19.3	(32)
(33)		10%	WB	5.5	11.0	27.5	34.5	42.6	60.9	63.4	59.6	47.3	34.7	16.1	11.3	(33)
(34)			MCD B	6.5	12.1	30.6	39.6	49.6	72.9	72.6	67.4	51.5	36.7	17.4	12.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.7	18.8	24.5	21.1	17.9	25.1	23.6	20.5	15.9	12.3	16.3	15.8	(35)
(36)			MCDBR	16.3	14.3	21.3	25.0	28.1	34.4	34.4	30.5	23.4	11.6	18.9	17.5	(36)
(37)			MCWBR	15.8	13.5	17.7	16.8	16.1	14.5	15.6	16.3	13.9	8.7	18.0	16.5	(37)
(38)		5% WB	MCDBR	16.2	14.3	19.8	21.2	26.6	28.8	25.8	24.8	20.0	10.0	19.0	16.2	(38)
(39)			MCWBR	15.7	13.3	16.8	15.1	16.4	13.3	13.6	14.4	12.7	8.0	18.3	15.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.230	0.281	0.314	0.342	0.361	0.376	0.427	0.407	0.340	0.287	0.210	0.215	(40)
(41)		taud		1.866	1.980	2.015	2.042	2.169	2.399	2.238	2.262	2.408	2.179	1.898	1.863	(41)
(42)		Ebn at Noon		90	187	236	259	267	267	246	239	235	189	98	31	(42)
(43)		Edh at Noon		10	23	33	41	40	33	37	33	23	18	9	4	(43)
(44)	All-Sky Solar Radiation	RadAvg		47	272	717	1287	1614	1765	1553	1038	619	251	82	19	(44)
(45)		RadStd		7	28	76	132	121	174	143	120	90	43	13	2	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page